

ADARSH GAURAV

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SUMMARY

CS undergraduate with hands-on experience in data science, machine learning, and predictive modeling. Skilled at building, evaluating, and deploying ML/deep learning models end-to-end — from feature engineering on structured and unstructured data to model architecture design and performance evaluation. Proficient in Python, SQL, and modern ML frameworks — looking to bring this analytical rigor to a data science role solving real-world problems at scale.

EXPERIENCE

Data Science Intern — CodSoft

June 2025 – July 2025

Remote

- Analyzed real-time datasets to surface patterns and trends, applying a problem-solving approach directly relevant to product analytics.
- Built and evaluated predictive models in Python, Pandas, and NumPy, turning raw data into clear, actionable recommendations.
- Designed data visualizations that made technical findings accessible to non-technical stakeholders, sharpening data storytelling skills.

Web Development Intern — CodeAlpha

Oct. 2024 – Nov. 2024

Remote

- Built and shipped responsive, user-facing web applications using HTML, Tailwind CSS, and JavaScript.
- Developed interactive UI components and real-time form validation, directly improving usability and cutting user error rates.
- Used structured debugging to ship reliable, production-ready features on schedule.

PROJECTS

RAG-based AI Teaching Assistant

Python, FFmpeg, OpenAI Whisper, BGE-M3, scikit-learn, NumPy, Llama 3.2, Ollama

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) pipeline combining speech-to-text, embedding-based retrieval, and LLM inference to power semantic Q&A over multi-hour lecture videos.
- Built a dense retrieval engine using BGE-M3 embeddings and cosine similarity search (scikit-learn/NumPy), tuning top-K retrieval to balance relevance and response latency.
- Automated transcription of lecture audio with OpenAI Whisper (large-v2) and FFmpeg, producing structured, time-stamped datasets for downstream model consumption.
- Integrated retrieved context with a locally hosted LLM (Llama 3.2 via Ollama) via custom prompt engineering to reduce hallucination and improve answer grounding.

Multimodal Deep Learning Framework for Heart Disease Prediction

CNN, BiLSTM, Transformer, ECG Signal Processing, Clinical Data Fusion, PyTorch, FastAPI, React.js, PostgreSQL, Scikit-learn

- Designed and trained a hybrid CNN–BiLSTM–Transformer architecture to fuse ECG signal data with structured clinical features for early heart disease prediction.
- Built a feature-fusion layer combining temporal (ECG) and tabular (clinical) representations, engineering both signal-processing and classical preprocessing pipelines.
- Achieved 96%+ classification accuracy and a 0.995 ROC-AUC, validated through cross-validation and held-out test evaluation.
- Deployed the trained model behind a FastAPI backend with a React.js interface and PostgreSQL persistence for real-time inference.

EDUCATION

Guru Ghasidas Vishwavidyalaya, A Central University, Bilaspur (C.G.)

Dec. 2022 – May 2026

B.Tech in Computer Science and Engineering, CGPA: 7.1/10

Central Public School, Bihar

April 2019 – March 2021

CBSE – Class XII, 70%

D.A.V. Public School, Bihar

April 2018 – March 2019

CBSE – Class X, 80.88%

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS

AI/Data: Machine Learning (SVM, predictive modeling), Pandas, NumPy, Data Analysis & Visualization

Tools: Git, Google Cloud Platform, VS Code, Visual Studio, Google Colab, Arduino IDE

LEADERSHIP & ACHIEVEMENTS

- Student Leader, Pregrad — Led campus tech initiatives, coordinating student teams across innovation programs.
- Deloitte Data Analytics Job Simulation — Delivered data-driven business recommendations in a simulated analytics engagement.